

CSCE 763: Digital Image Processing

Homework #2

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Due: 1:15 pm EST, Tuesday, Feb 10

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Problem 1: Median Operator Nonlinearity (10 pts)

The median ζ , of a set of numbers is such that half the values in the set are below ζ , and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S , is nonlinear.

Hints: an operator H is linear if

$$H[a_1f_1(x, y) + a_2f_2(x, y)] = a_1H[f_1(x, y)] + a_2H[f_2(x, y)]$$

Step 1: State the Linearity Condition

An operator H is linear if and only if the **superposition property** holds for all images f_1, f_2 and all scalars a_1, a_2 :

$$H[a_1f_1 + a_2f_2] = a_1H[f_1] + a_2H[f_2]$$

To show H is nonlinear, it suffices to find a single counterexample where this equality fails. We set $a_1 = a_2 = 1$ and test the additivity property over a 1×3 subimage region S .

Step 2: Define the Counterexample

Choose two images defined by their pixel values over S :

$$f_1 = \{2, 3, 10\}, \quad f_2 = \{10, 1, 2\}$$

Step 3: Compute the Right-Hand Side

$$H[f_1] + H[f_2] = \text{median}\{2, 3, 10\} + \text{median}\{10, 1, 2\} = 3 + 2 = 5$$

Step 4: Compute the Left-Hand Side

$$f_1 + f_2 = \{2 + 10, 3 + 1, 10 + 2\} = \{12, 4, 12\}$$

$$H[f_1 + f_2] = \text{median}\{12, 4, 12\} = 12$$

Step 5: Conclude

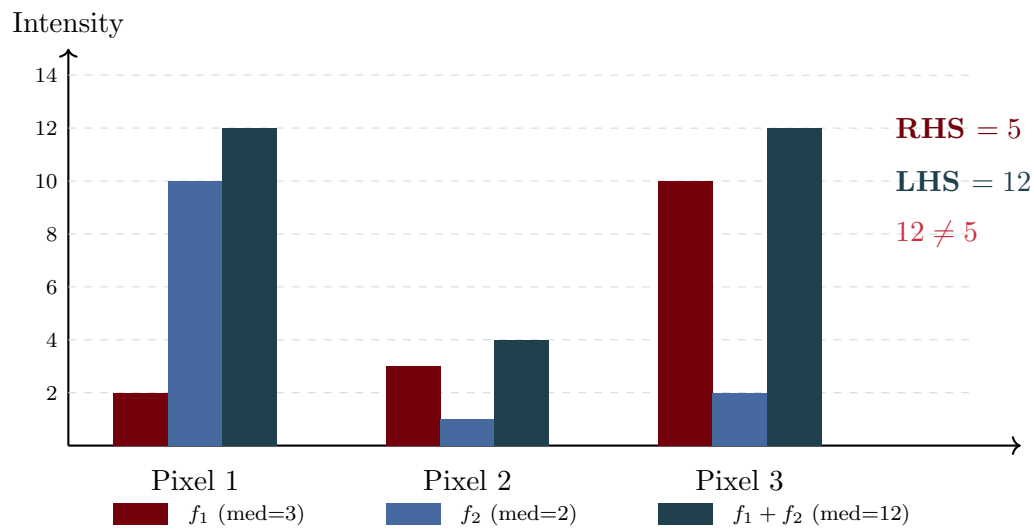
Since

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2]$$

the superposition property is violated. Therefore, the median operator is **nonlinear**.

■

Visualization: Counterexample Bar Chart



Results

The median operator is **nonlinear**. The counterexample $f_1 = \{2, 3, 10\}$, $f_2 = \{10, 1, 2\}$ yields:

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2] \quad \blacksquare$$

Problem 2: Image Subtraction for Defect Detection (20 pts)

Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance.

To demonstrate these factors concretely, I applied various perturbations to a “golden” reference image of an industrial product (a pharmaceutical tablet) and computed the resulting difference images. Figure 1 shows the golden reference image used for this analysis.



Figure 1: Golden reference image of a pharmaceutical tablet with “FF” marking, used as the baseline for defect detection analysis.

Factor 1: Illumination Intensity

Changes in overall lighting intensity cause uniform brightness shifts between images. Even small changes (± 40 intensity levels) produce significant difference signals across the entire image, creating false positives that completely mask actual defects.

Practical requirement: Fixed, calibrated light sources in enclosed inspection chambers.

Factor 2: Spatial Translation

If the product shifts position between captures—even by a few pixels—edges and high-contrast features produce large differences at boundaries. A 5-pixel shift creates false positives along all edges of the product.

Practical requirement: Mechanical fixtures, conveyor registration, or sub-pixel image alignment algorithms.

Factor 3: Rotational Misalignment

Angular misalignment (even 2°) causes the entire image to differ at every edge location. The difference pattern forms an arc around the object perimeter, which could easily be confused with circular defects.

Practical requirement: Rotational alignment fixtures or automated rotation correction.

Factor 4: Scale/Zoom Variation

Changes in camera-to-object distance or zoom settings alter the apparent size of the product. A 3% scale change creates significant differences throughout the image, especially near edges.

Practical requirement: Fixed focal length, fixed working distance, or scale normalization preprocessing.

Factor 5: Sensor Noise

Random sensor noise ($\sigma = 20$ Gaussian) creates speckle-like differences uniformly distributed across the image. While individual noise pixels may be below threshold, accumulated noise raises the detection floor.

Practical requirement: Low-noise sensors, temporal averaging, or adaptive thresholding.

Factor 6: Focus Blur

If the test image is slightly out of focus compared to the golden image, edge sharpness differs. Blurring reduces edge contrast, creating differences where edges become softer.

Practical requirement: Auto-focus locks, fixed-focus telecentric lenses, or depth-of-field control.

Factor 7: Non-uniform Illumination (Vignetting)

Optical vignetting or uneven light distribution causes the image corners/edges to be darker than the center. This creates a characteristic radial difference pattern.

Practical requirement: Flat-field correction, uniform light sources, or calibrated vignetting correction.

Factor 8: Contrast/Gain Variation

Changes in camera gain or lighting contrast affect the dynamic range of the captured image. A 30% contrast change produces differences proportional to local intensity.

Practical requirement: Fixed camera settings, calibrated exposure, auto-gain disabled.

Factor 9: Color Temperature / White Balance

Shifts in lighting color temperature (e.g., warm vs. cool) or white balance settings cause color channel differences. In grayscale conversion, this manifests as intensity shifts.

Practical requirement: Fixed white balance, controlled spectrum lighting (LED with fixed CCT).

Factor 10: Perspective Distortion

Changes in viewing angle create trapezoidal distortions. Even a 5% perspective shift causes significant geometric differences, especially near image edges.

Practical requirement: Telecentric optics, fixed camera mount, perspective calibration.

Factor 11: Exposure Variation

Changes in exposure time (measured in EV stops) shift the overall image brightness non-linearly. Overexposure saturates bright regions while underexposure loses shadow detail.

Practical requirement: Fixed shutter speed, controlled ambient light, exposure calibration.

Factor 12: Directional Shadows

Shadows cast by external objects or non-uniform lighting create localized intensity variations. These appear as gradient differences across the image.

Practical requirement: Diffuse lighting, light tents, shadow-free illumination geometry.

Figure 2 shows a summary of six key failure modes applied to the golden image. Each column shows: (1) the perturbed test image, (2) the resulting difference image (brighter = larger difference), and (3) the false positives detected at a fixed threshold. Notice how environmental variations that leave the product physically unchanged still trigger significant detection artifacts.

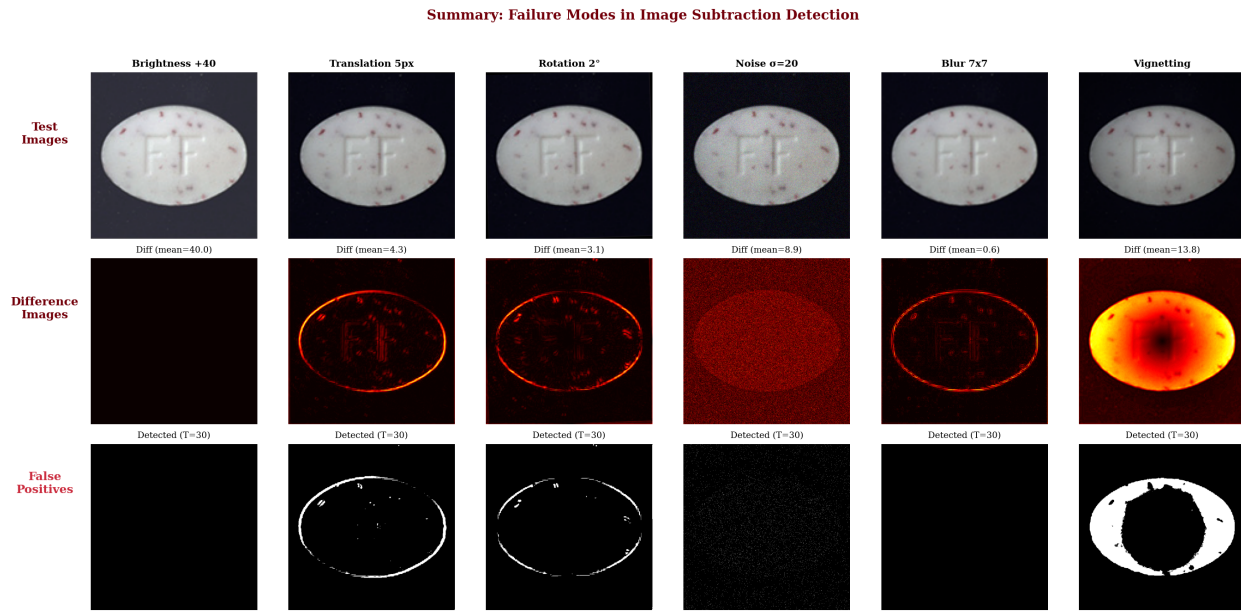


Figure 2: Summary of failure modes in image subtraction detection. Row 1: Test images with various perturbations. Row 2: Difference images (hotter = larger difference). Row 3: Detected regions at threshold $T = 30$. All perturbations produce false positives despite no actual defects.

Figure 3 quantitatively compares all twelve failure modes by mean absolute difference. The key insight is that **illumination-related factors dominate**: brightness changes and exposure variations cause the largest false positive signals. Geometric factors (translation, rotation, scale) cause smaller mean differences but concentrate at edges where defects are often located. Critically, a true defect (missing component) has a relatively **small** mean difference because it affects only a localized region—demonstrating how failure modes can mask real defects.

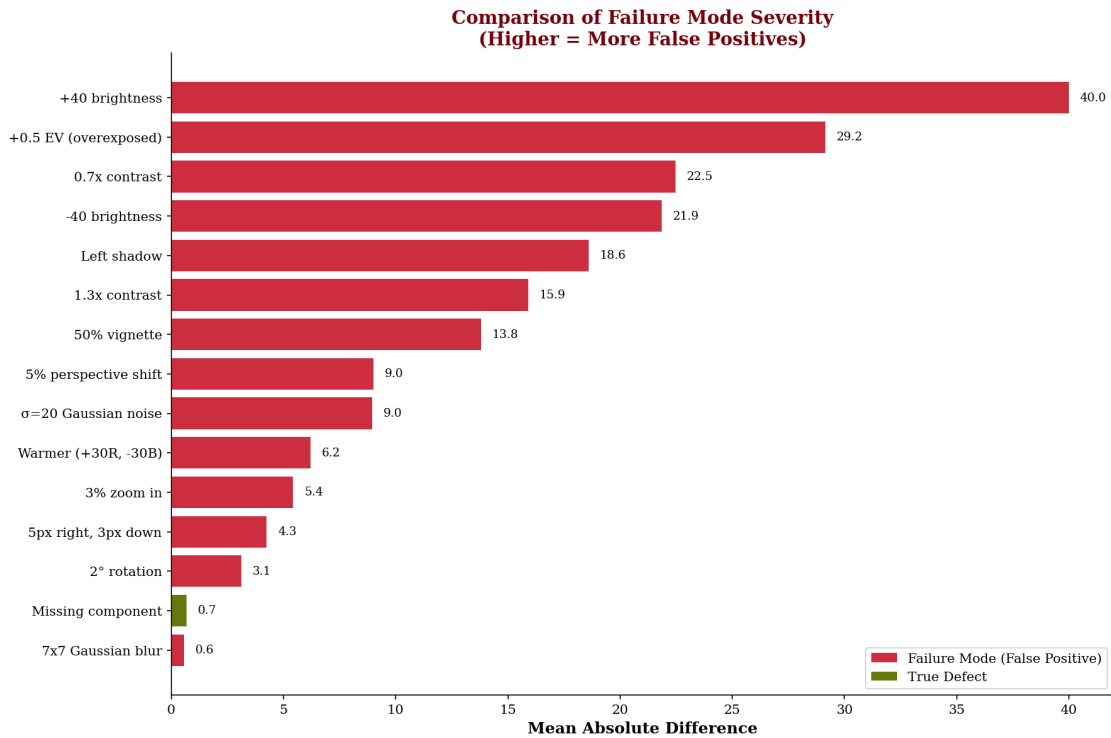


Figure 3: Comparison of failure mode severity measured by mean absolute difference. Higher values indicate more false positives. Note that the true defect (missing component, green) has a *lower* mean difference than most failure modes, illustrating how environmental variations can mask genuine defects.

Results

Major factors affecting image subtraction defect detection:

Primary Factors (Must Control):

1. **Illumination consistency:** Intensity, direction, uniformity, and color temperature must be identical.
2. **Spatial registration:** Position, orientation, and scale must match exactly.

Secondary Factors (Should Control):

3. Sensor noise (requires thresholding)
4. Focus/sharpness consistency
5. Camera gain and exposure settings
6. White balance / color calibration
7. Perspective and optical distortion
8. Shadow elimination

Key insight: Without controlling these factors, the mean difference from environmental variations can *exceed* the difference caused by actual defects, making reliable detection impossible.

Problem 3: Histogram Equalization Idempotency (20 pts)

Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass.

Step 1: Define the Equalization Transform

Let the original image have intensity levels $r \in [0, L - 1]$ with PDF $p_r(r)$ and CDF $P_r(r)$. The histogram equalization transformation is:

$$s = T(r) = (L - 1) P_r(r) = (L - 1) \int_0^r p_r(w) dw$$

Step 2: Show the Equalized Image Is Uniform

A fundamental result from probability theory: if $s = T(r) = (L - 1) P_r(r)$, then s is **uniformly distributed** on $[0, L - 1]$:

$$p_s(s) = \frac{1}{L - 1}, \quad 0 \leq s \leq L - 1$$

Step 3: Apply Equalization Again

Apply histogram equalization a second time to the equalized image:

$$v = T_2(s) = (L - 1) P_s(s) = (L - 1) \int_0^s p_s(w) dw$$

Since $p_s(s) = \frac{1}{L - 1}$, the CDF is:

$$P_s(s) = \int_0^s \frac{1}{L - 1} dw = \frac{s}{L - 1}$$

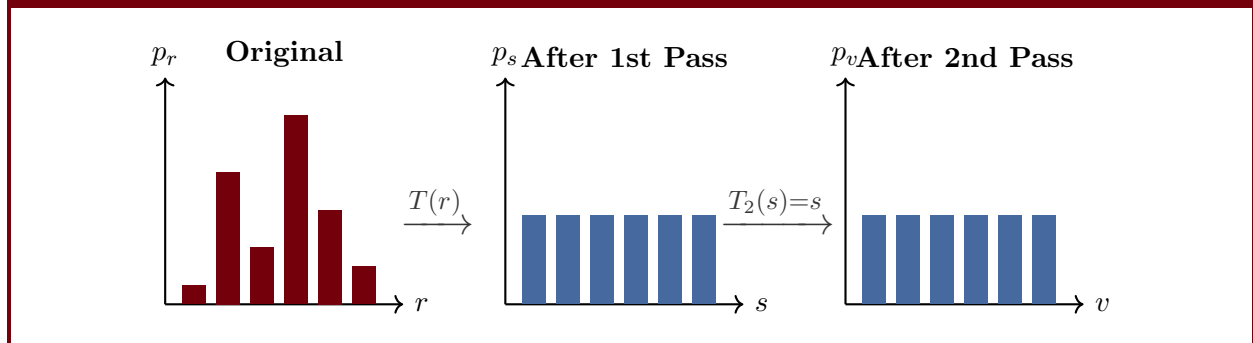
Step 4: Simplify to Identity

Substituting:

$$v = T_2(s) = (L - 1) \cdot \frac{s}{L - 1} = s$$

The second equalization is the **identity transformation**: $v = s$. The output is unchanged. ■

Visualization: Histogram Through Two Equalization Passes



Results

After the first histogram equalization, the output has a uniform distribution. Applying histogram equalization to a uniformly distributed image yields the identity mapping $v = s$. Therefore, the second pass produces exactly the same result as the first. ■

Problem 4: Histogram Matching (30 pts)

Suppose that a 4-bit image ($L = 16$) of size 64×64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation?

r_k	n_k
0	16
1	0
2	512
3	32
4	64
5	128
6	64
7	1024
8	512
9	512
10	16
11	1024
12	0
13	128
14	0
15	64

Table A

z_q	n_q
0	64
1	128
2	512
3	512
4	1024
5	1024
6	512
7	128
8	128
9	64
10	0
11	0
12	0
13	0
14	0
15	0

Table B
Step 1: Setup

Total pixels: $N = 64 \times 64 = 4096$, intensity levels: $L = 16$, so $L - 1 = 15$.
The equalization transform is:

$$s_k = \text{round} \left(15 \cdot \sum_{j=0}^k \frac{n_j}{4096} \right)$$

Step 2: Cumulative Histogram and s_k Values from Table A

r_k	n_k	$\sum_{j=0}^k n_j$	$s_k = \text{round}\left(\frac{15 \cdot \sum_{j=0}^k n_j}{4096}\right)$
0	16	16	$\text{round}(0.059) = 0$
1	0	16	$\text{round}(0.059) = 0$
2	512	528	$\text{round}(1.934) = 2$
3	32	560	$\text{round}(2.051) = 2$
4	64	624	$\text{round}(2.285) = 2$
5	128	752	$\text{round}(2.754) = 3$
6	64	816	$\text{round}(2.988) = 3$
7	1024	1840	$\text{round}(6.738) = 7$
8	512	2352	$\text{round}(8.613) = 9$
9	512	2864	$\text{round}(10.488) = 10$
10	16	2880	$\text{round}(10.547) = 11$
11	1024	3904	$\text{round}(14.297) = 14$
12	0	3904	$\text{round}(14.297) = 14$
13	128	4032	$\text{round}(14.766) = 15$
14	0	4032	$\text{round}(14.766) = 15$
15	64	4096	$\text{round}(15.000) = 15$

Step 3: Cumulative Histogram and $G(z_q)$ Values from Table B

z_q	n_q	$\sum_{j=0}^q n_j$	$G(z_q) = \text{round}\left(\frac{15 \cdot \sum_{j=0}^q n_j}{4096}\right)$
0	64	64	$\text{round}(0.234) = 0$
1	128	192	$\text{round}(0.703) = 1$
2	512	704	$\text{round}(2.578) = 3$
3	512	1216	$\text{round}(4.453) = 4$
4	1024	2240	$\text{round}(8.203) = 8$
5	1024	3264	$\text{round}(11.953) = 12$
6	512	3776	$\text{round}(13.828) = 14$
7	128	3904	$\text{round}(14.297) = 14$
8	128	4032	$\text{round}(14.766) = 15$
9	64	4096	$\text{round}(15.000) = 15$
10	0	4096	15
11	0	4096	15
12	0	4096	15
13	0	4096	15
14	0	4096	15
15	0	4096	15

Step 4: Inverse Mapping $s_k \rightarrow z_q$

For each value of s_k , find the smallest z_q such that $G(z_q) \geq s_k$:

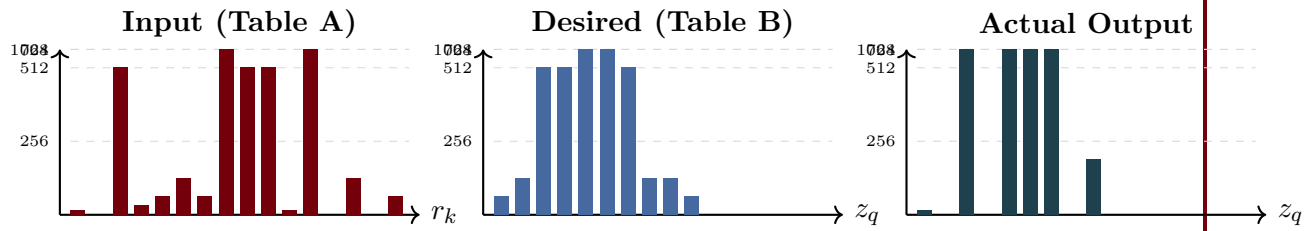
r_k	s_k	Smallest z_q with $G(z_q) \geq s_k$	z_q
0	0	$G(0) = 0 \geq 0$	0
1	0	$G(0) = 0 \geq 0$	0
2	2	$G(2) = 3 \geq 2$	2
3	2	$G(2) = 3 \geq 2$	2
4	2	$G(2) = 3 \geq 2$	2
5	3	$G(2) = 3 \geq 3$	2
6	3	$G(2) = 3 \geq 3$	2
7	7	$G(4) = 8 \geq 7$	4
8	9	$G(5) = 12 \geq 9$	5
9	10	$G(5) = 12 \geq 10$	5
10	11	$G(5) = 12 \geq 11$	5
11	14	$G(6) = 14 \geq 14$	6
12	14	$G(6) = 14 \geq 14$	6
13	15	$G(8) = 15 \geq 15$	8
14	15	$G(8) = 15 \geq 15$	8
15	15	$G(8) = 15 \geq 15$	8

Step 5: Build the Output Histogram

Using the mapping $r_k \rightarrow z_q$ and the pixel counts n_k from Table A:

Output z_q	Mapped from r_k	Pixel count
0	r_0, r_1	$16 + 0 = 16$
1	—	0
2	r_2, r_3, r_4, r_5, r_6	$512 + 32 + 64 + 128 + 64 = 800$
3	—	0
4	r_7	1024
5	r_8, r_9, r_{10}	$512 + 512 + 16 = 1040$
6	r_{11}, r_{12}	$1024 + 0 = 1024$
7	—	0
8	r_{13}, r_{14}, r_{15}	$128 + 0 + 64 = 192$
9–15	—	0 each
Total:		$16 + 800 + 1024 + 1040 + 1024 + 192 = 4096 \checkmark$

Visualization: Input



Results

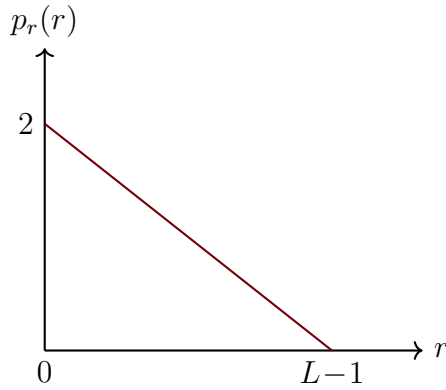
The actual output histogram after histogram matching is:

z_q	n_q^{out}
0	16
1	0
2	800
3	0
4	1024
5	1040
6	1024
7	0
8	192
9	0
10	0
11	0
12	0
13	0
14	0
15	0

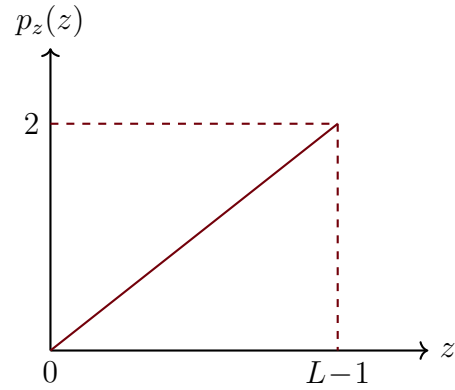
The output histogram approximates the desired distribution but is not identical to it, due to the discrete nature of the intensity levels and the many-to-one mapping inherent in histogram matching.

Problem 5: Continuous Intensity Transformation (20 pts)

An image with intensities in the range $[0, L - 1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.



(a)



(b)

Step 1: Write $p_r(r)$ with Normalization Verification

Let $W = L - 1$. From Figure (a), $p_r(r)$ is a line from $(0, \frac{2}{W})$ to $(W, 0)$:

$$p_r(r) = \frac{2}{W} \left(1 - \frac{r}{W} \right) = \frac{2(W - r)}{W^2}, \quad 0 \leq r \leq W$$

$$\text{Verification: } \int_0^W \frac{2(W - r)}{W^2} dr = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right]_0^W = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 2: Write $p_z(z)$ with Normalization Verification

From Figure (b), $p_z(z)$ is a line from $(0, 0)$ to $(W, \frac{2}{W})$:

$$p_z(z) = \frac{2z}{W^2}, \quad 0 \leq z \leq W$$

$$\text{Verification: } \int_0^W \frac{2z}{W^2} dz = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 3: Compute CDF $T(r)$

$$T(r) = \int_0^r p_r(w) dw = \int_0^r \frac{2(W-w)}{W^2} dw = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right] = \frac{2Wr - r^2}{W^2}$$

Step 4: Compute CDF $G(z)$

$$G(z) = \int_0^z p_z(w) dw = \int_0^z \frac{2w}{W^2} dw = \frac{z^2}{W^2}$$

Step 5: Set $G(z)$

$$\begin{aligned} \frac{z^2}{W^2} &= \frac{2Wr - r^2}{W^2} \\ z^2 &= 2Wr - r^2 = r(2W - r) \\ z &= \sqrt{r(2(L-1) - r)} \end{aligned}$$

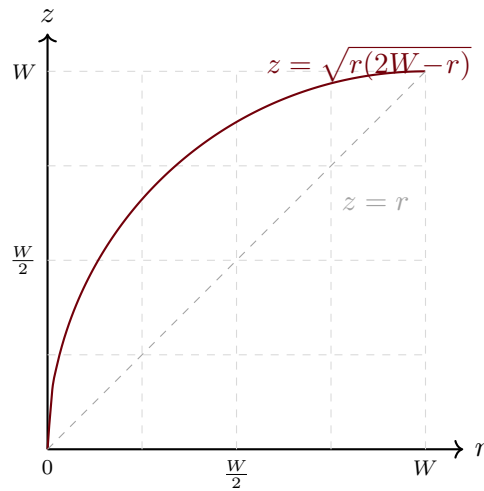
where we take the positive root since $0 \leq r \leq W$ ensures the argument is non-negative.

Step 6: Verify Boundary Conditions

- $f(0) = \sqrt{0 \cdot (2W - 0)} = 0 \checkmark$
- $f(W) = \sqrt{W(2W - W)} = \sqrt{W^2} = W = L - 1 \checkmark$

The transformation maps $[0, L - 1] \rightarrow [0, L - 1]$ as required.

Visualization: Transformation z



The curve is a quarter-circle arc from $(0,0)$ to (W,W) , lying above the identity line, indicating the transformation boosts lower intensities.

Results

The transformation that maps the input PDF $p_r(r)$ to the desired output PDF $p_z(z)$ is:

$$z = f(r) = \sqrt{r(2(L-1) - r)}, \quad 0 \leq r \leq L-1$$

This is a quarter-circle arc from $(0,0)$ to $(L-1, L-1)$. ■